

Lab12 Report

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1. Core to IO boundary :

Innovus(TM) Implementation System 20.15 - /RAID2/COURSE/iclab/iclab047/Lab12/Exercise/05_APR/CHIP.inn.dat - CHIP — ee21

Specify Floorplan — ee21

Basic Advanced

Design Dimensions

Specify By: ☒ Size ☐ Die/IO/Core Coordinates

☒ Core Size by: ☒ Aspect Ratio: Ratio (H/W): 434688902

☒ Core Utilization: 1.0

☐ Cell Utilization: 1.0

☐ Dimension: Width: 2264.24

Height: 2262.96

☐ Die Size by: Width: 2745.36

Height: 2745.36

Core Margins by: ☒ Core to IO Boundary

☐ Core to Die Boundary

Core to Left: 100.44 Core to Top: 101.48

Core to Right: 100.44 Core to Bottom: 100.68

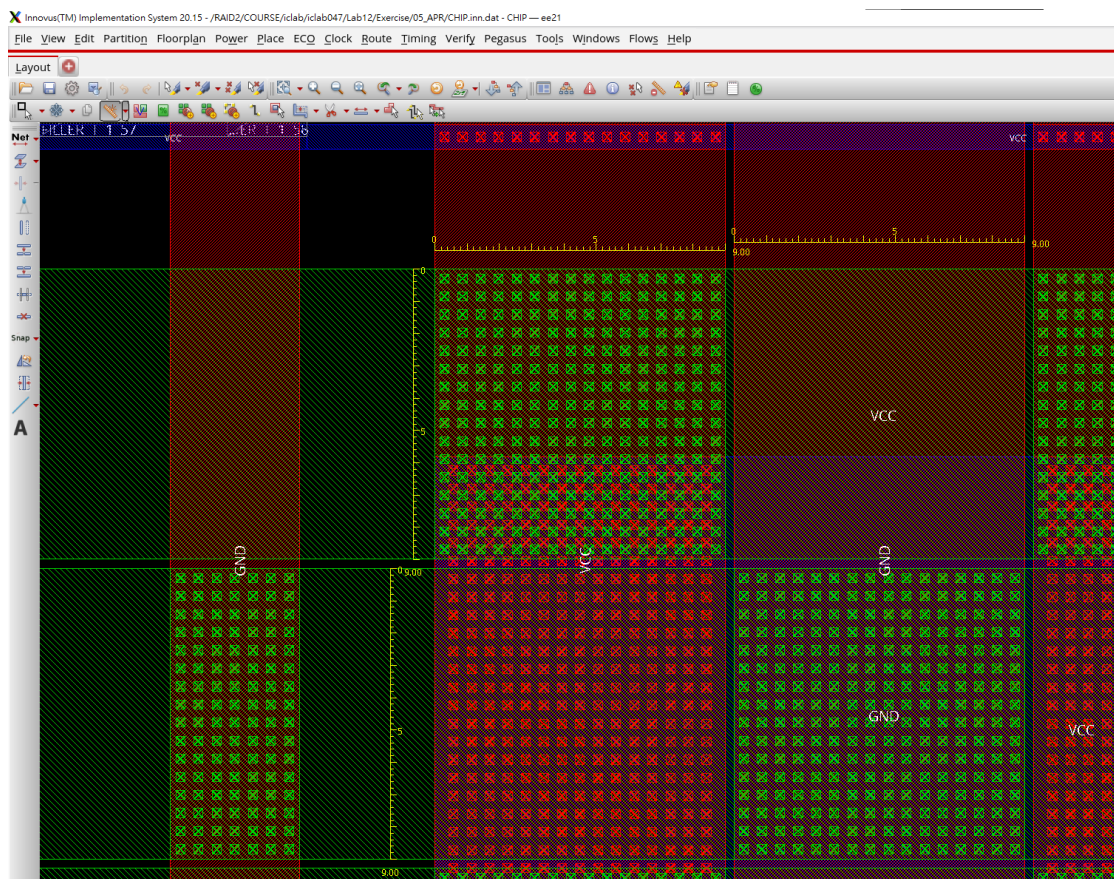
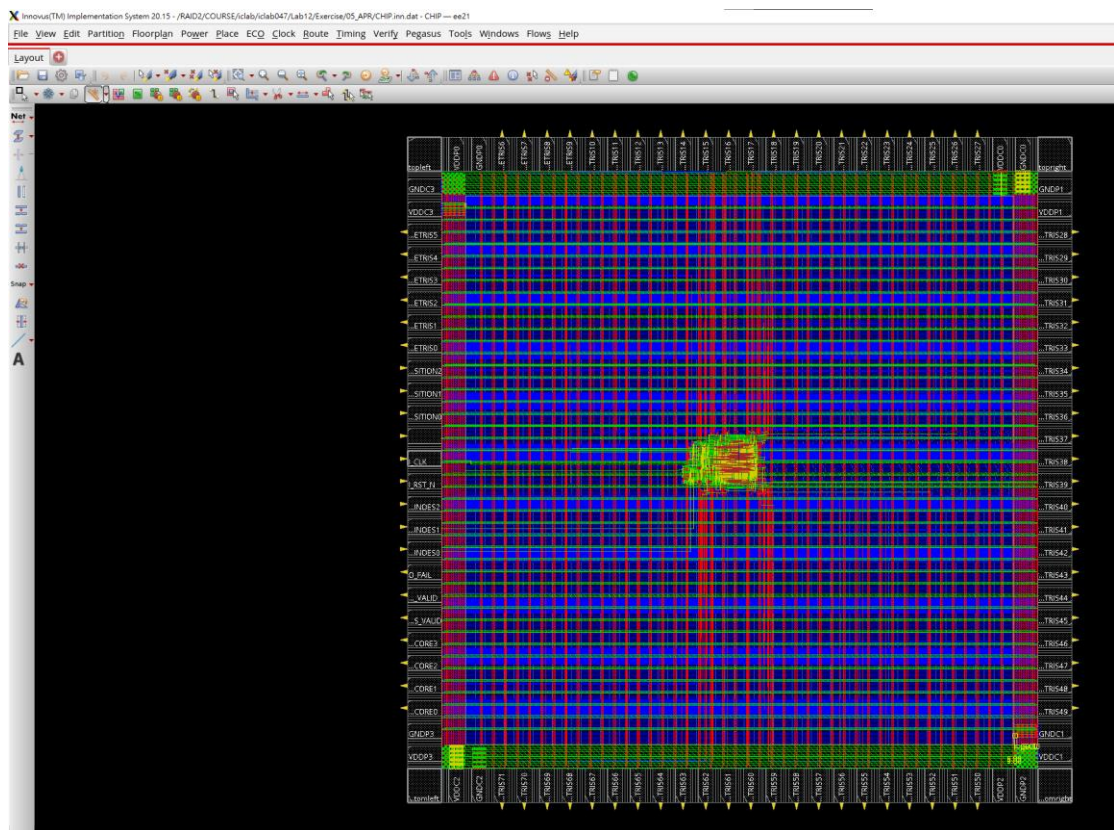
Die Size Calculation Use: ☐ Max IO Height ☒ Min IO Height

Floorplan Origin at: ☒ Lower Left Corner ☐ Center

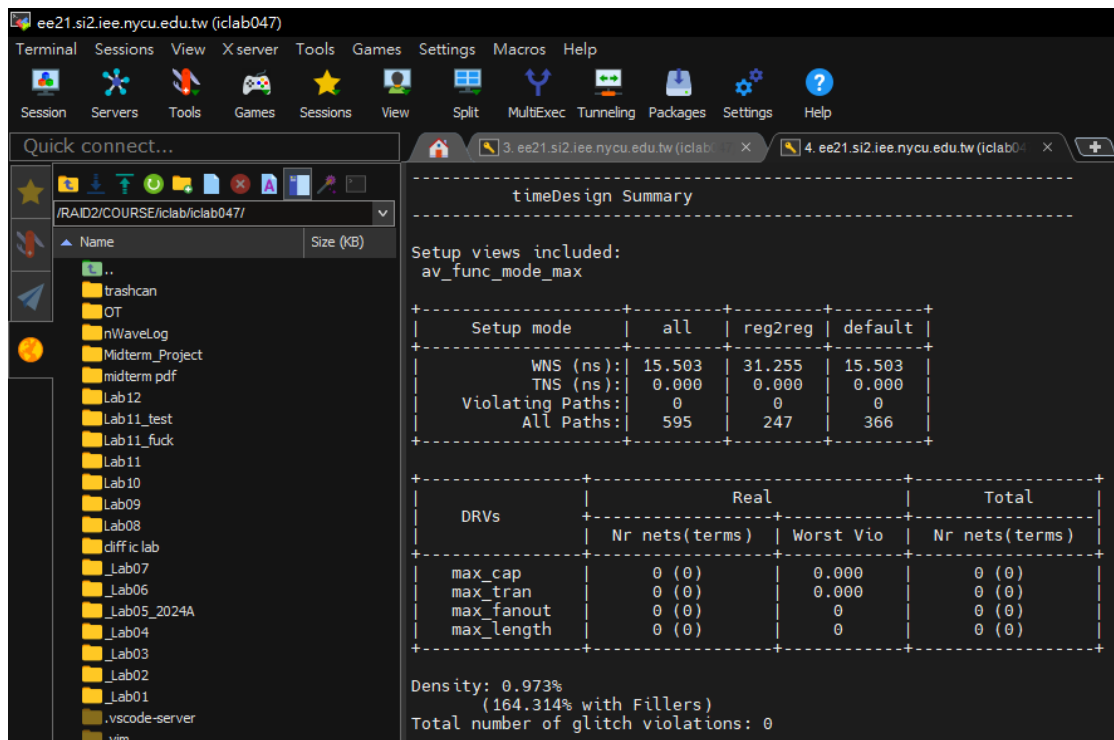
Unit: Micron

OK Apply Cancel Help

2. Core Ring :



3. Post-Route setup time analysis :



The screenshot shows a terminal window with a dark theme. The title bar indicates the user is logged in as 'ee21.si2.iee.nycu.edu.tw (iclab047)'. The menu bar includes Terminal, Sessions, View, X server, Tools, Games, Settings, Macros, and Help. The toolbar contains icons for Session, Servers, Tools, Games, Sessions, View, Split, MultiExec, Tunneling, Packages, Settings, and Help. The left sidebar shows a file explorer with a 'Quick connect...' section and a list of files and folders, including 'trashcan', 'OT', 'nWaveLog', 'Midterm_Project', 'midterm.pdf', 'Lab12', 'Lab11_test', 'Lab11_fuck', 'Lab11', 'Lab10', 'Lab09', 'Lab08', 'diff ic lab', '_Lab07', '_Lab06', '_Lab05_2024A', '_Lab04', '_Lab03', '_Lab02', '_Lab01', '.vscode-server', and '.vim'. The main terminal area displays a 'timeDesign Summary' report.

```
timeDesign Summary
```

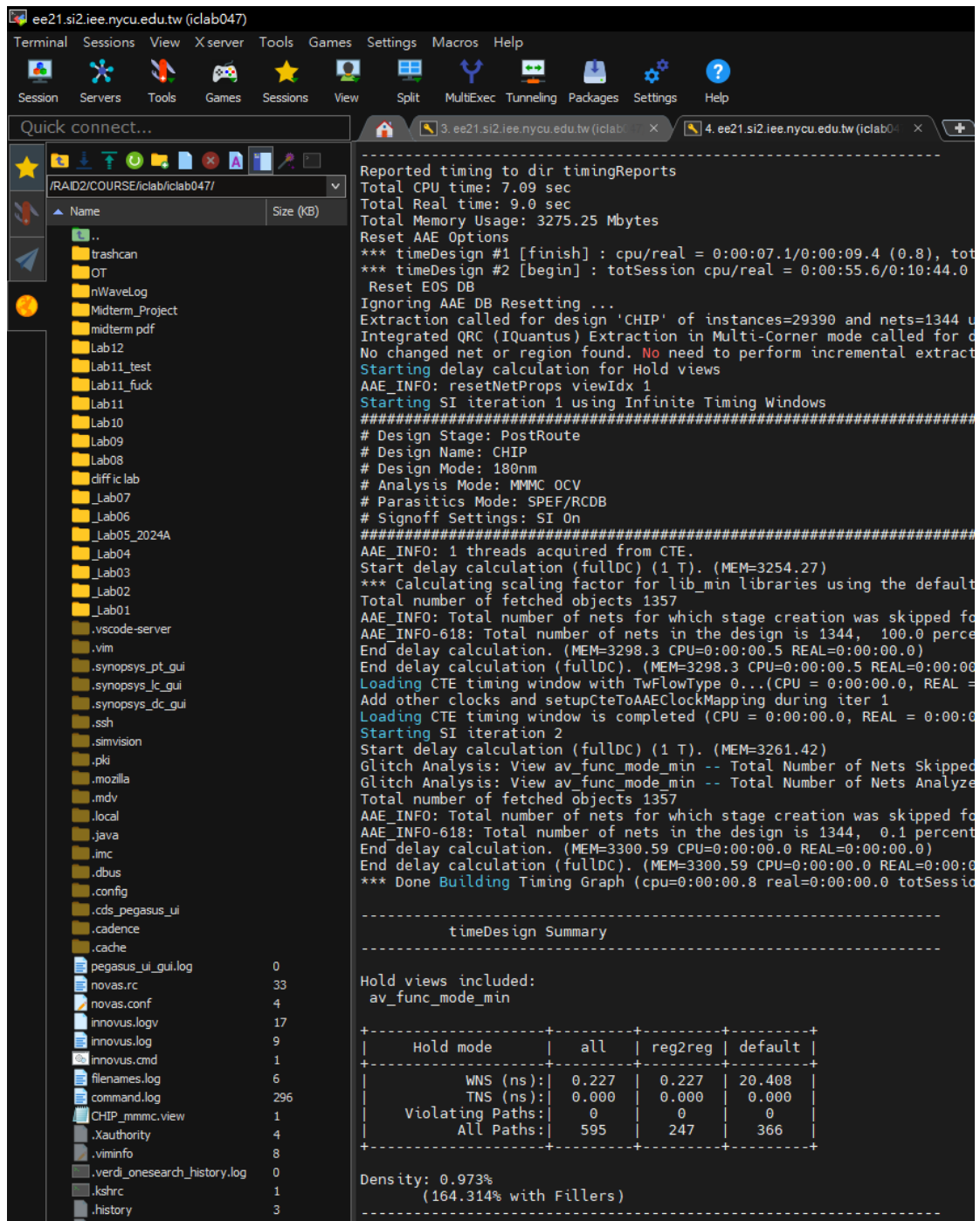
Setup views included:
av_func_mode_max

Setup mode	all	reg2reg	default
WNS (ns):	15.503	31.255	15.503
TNS (ns):	0.000	0.000	0.000
Violating Paths:	0	0	0
All Paths:	595	247	366

DRVs	Real		Total
	Nr nets(terms)	Worst Vio	Nr nets(terms)
max_cap	0 (0)	0.000	0 (0)
max_tran	0 (0)	0.000	0 (0)
max_fanout	0 (0)	0	0 (0)
max_length	0 (0)	0	0 (0)

Density: 0.973%
(164.314% with Fillers)
Total number of glitch violations: 0

4. Post-Route hold time analysis :



```
ee21.si2.tee.nycu.edu.tw (iclab047)
Terminal Sessions View X server Tools Games Settings Macros Help
Session Servers Tools Games Sessions View Split MultiExec Tunneling Packages Settings Help

Quick connect...
/RAID2/COURSE/iclab/iclab047/
Name Size (kB)
trashcan
OT
nWaveLog
Midterm_Project
midterm.pdf
Lab12
Lab11_test
Lab11_fuck
Lab11
Lab10
Lab09
Lab08
difficlab
_Lab07
_Lab06
_Lab05_2024A
_Lab04
_Lab03
_Lab02
_Lab01
.vscode-server
.vim
.synopsys_pt_gui
.synopsys_lc_gui
.synopsys_dc_gui
.ssh
.simvision
.pki
.mozilla
.mdv
.local
.java
.imc
.dbus
.config
.cds_pegasus_ui
.cadence
.cache
pegasus_ui_gui.log 0
novas.rc 33
novas.conf 4
innovus.logv 17
innovus.log 9
innovus.cmd 1
filenames.log 6
command.log 296
CHIP_mmmc.view 1
.Xauthority 4
.viminfo 8
.verdi_oneseach_history.log 0
.kshrc 1
.history 3

Reported timing to dir timingReports
Total CPU time: 7.09 sec
Total Real time: 9.0 sec
Total Memory Usage: 3275.25 Mbytes
Reset AAE Options
*** timeDesign #1 [finish] : cpu/real = 0:00:07.1/0:00:09.4 (0.8), tot
*** timeDesign #2 [begin] : totSession cpu/real = 0:00:55.6/0:10:44.0
Reset EOS DB
Ignoring AAE DB Resetting ...
Extraction called for design 'CHIP' of instances=29390 and nets=1344 u
Integrated QRC (fQuantus) Extraction in Multi-Corner mode called for d
No changed net or region found. No need to perform incremental extract
Starting delay calculation for Hold views
AAE_INFO: resetNetProps viewIdx 1
Starting SI iteration 1 using Infinite Timing Windows
#####
# Design Stage: PostRoute
# Design Name: CHIP
# Design Mode: 180nm
# Analysis Mode: MMMC OCV
# Parasitics Mode: SPEF/RADB
# Signoff Settings: SI On
#####
AAE_INFO: 1 threads acquired from CTE.
Start delay calculation (fullDC) (1 T). (MEM=3254.27)
*** Calculating scaling factor for lib_min libraries using the default
Total number of fetched objects 1357
AAE_INFO: Total number of nets for which stage creation was skipped fo
AAE_INFO-618: Total number of nets in the design is 1344, 100.0 perce
End delay calculation. (MEM=3298.3 CPU=0:00:00.5 REAL=0:00:00.0)
End delay calculation (fullDC). (MEM=3298.3 CPU=0:00:00.5 REAL=0:00:00
Loading CTE timing window with TwFlowType 0...(CPU = 0:00:00.0, REAL =
Add other clocks and setupCteToAAEClockMapping during iter 1
Loading CTE timing window is completed (CPU = 0:00:00.0, REAL = 0:00:0
Starting SI iteration 2
Start delay calculation (fullDC) (1 T). (MEM=3261.42)
Glitch Analysis: View av_func_mode_min -- Total Number of Nets Skipped
Glitch Analysis: View av_func_mode_min -- Total Number of Nets Analyze
Total number of fetched Objects 1357
AAE_INFO: Total number of nets for which stage creation was skipped fo
AAE_INFO-618: Total number of nets in the design is 1344, 0.1 percent
End delay calculation. (MEM=3300.59 CPU=0:00:00.0 REAL=0:00:00.0)
End delay calculation (fullDC). (MEM=3300.59 CPU=0:00:00.0 REAL=0:00:0
*** Done Building Timing Graph (cpu=0:00:00.8 real=0:00:00.0 totSessio

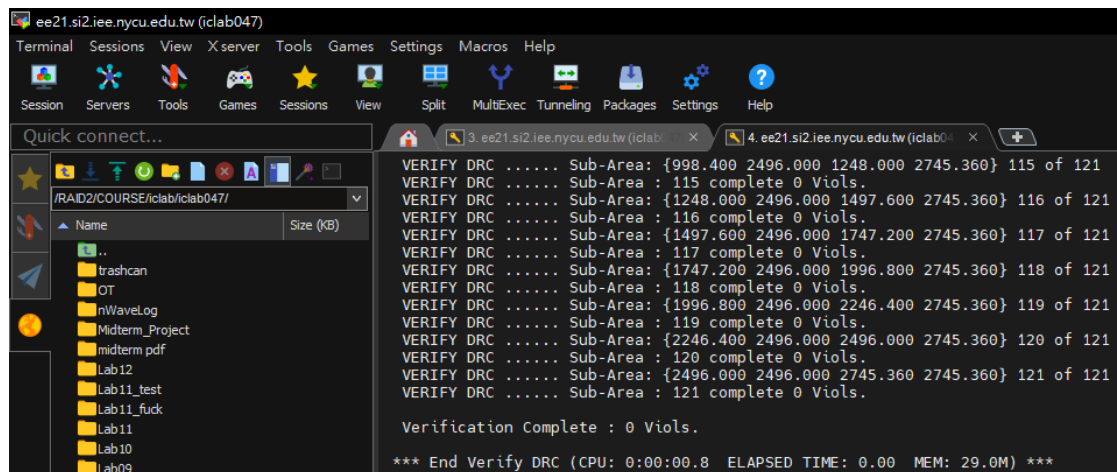
timeDesign Summary

Hold views included:
av_func_mode_min

+-----+-----+-----+-----+
| Hold mode | all | reg2reg | default |
+-----+-----+-----+-----+
| WNS (ns): | 0.227 | 0.227 | 20.408 |
| TNS (ns): | 0.000 | 0.000 | 0.000 |
| Violating Paths: | 0 | 0 | 0 |
| All Paths: | 595 | 247 | 366 |
+-----+-----+-----+-----+

Density: 0.973%
(164.314% with Fillers)
```

5. DRC result :



```
ee21.si2.tee.nycu.edu.tw (iclab047)
Terminal Sessions View X server Tools Games Settings Macros Help
Session Servers Tools Games Sessions View Split MultiExec Tunneling Packages Settings Help

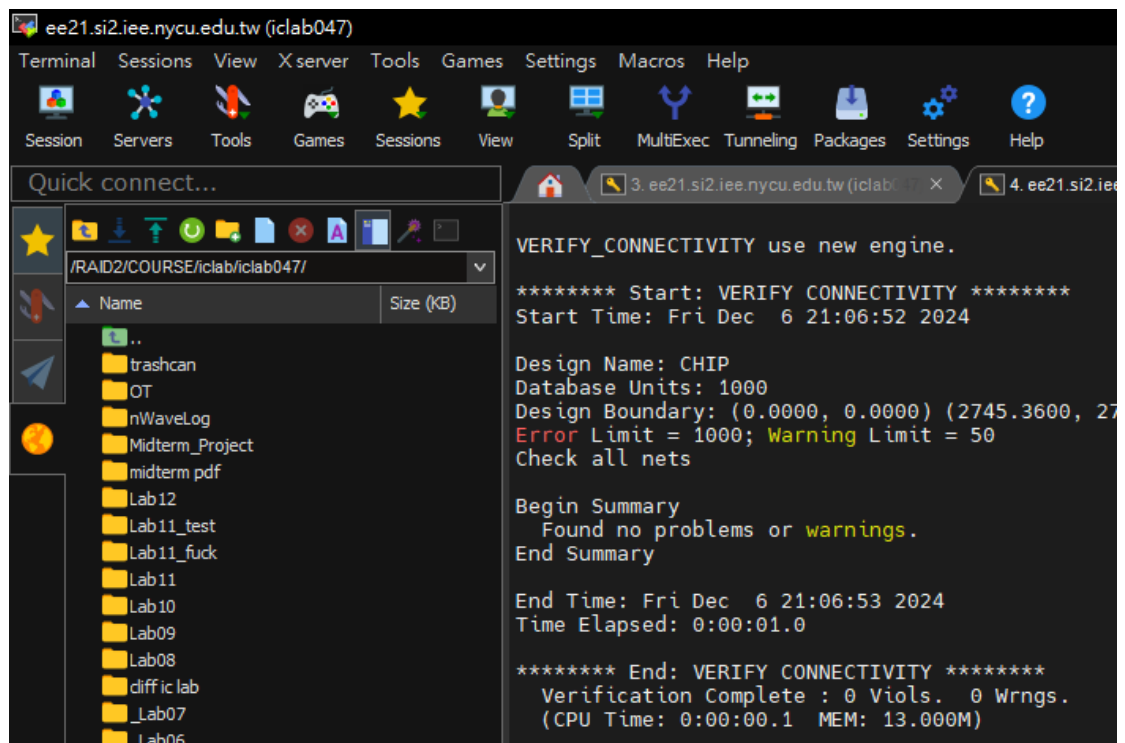
Quick connect...
/RAID2/COURSE/iclab/iclab047/
Name Size (KB)
trashcan
OT
nWaveLog
Midterm_Project
midterm.pdf
Lab12
Lab11_test
Lab11_fuck
Lab11
Lab10
Lab09

VERIFY DRC ..... Sub-Area: {998.400 2496.000 1248.000 2745.360} 115 of 121
VERIFY DRC ..... Sub-Area: : 115 complete 0 Viols.
VERIFY DRC ..... Sub-Area: {1248.000 2496.000 1497.600 2745.360} 116 of 121
VERIFY DRC ..... Sub-Area: : 116 complete 0 Viols.
VERIFY DRC ..... Sub-Area: {1497.600 2496.000 1747.200 2745.360} 117 of 121
VERIFY DRC ..... Sub-Area: : 117 complete 0 Viols.
VERIFY DRC ..... Sub-Area: {1747.200 2496.000 1996.800 2745.360} 118 of 121
VERIFY DRC ..... Sub-Area: : 118 complete 0 Viols.
VERIFY DRC ..... Sub-Area: {1996.800 2496.000 2246.400 2745.360} 119 of 121
VERIFY DRC ..... Sub-Area: : 119 complete 0 Viols.
VERIFY DRC ..... Sub-Area: {2246.400 2496.000 2496.000 2745.360} 120 of 121
VERIFY DRC ..... Sub-Area: : 120 complete 0 Viols.
VERIFY DRC ..... Sub-Area: {2496.000 2496.000 2745.360 2745.360} 121 of 121
VERIFY DRC ..... Sub-Area: : 121 complete 0 Viols.

Verification Complete : 0 Viols.

*** End Verify DRC (CPU: 0:00:00.8 ELAPSED TIME: 0.00 MEM: 29.0M) ***
```

6. LVS result :



```
ee21.si2.tee.nycu.edu.tw (iclab047)
Terminal Sessions View X server Tools Games Settings Macros Help
Session Servers Tools Games Sessions View Split MultiExec Tunneling Packages Settings Help

Quick connect...
/RAID2/COURSE/iclab/iclab047/
Name Size (KB)
trashcan
OT
nWaveLog
Midterm_Project
midterm.pdf
Lab12
Lab11_test
Lab11_fuck
Lab11
Lab10
Lab09
Lab08
diffic lab
_Lab07
Lab06

VERIFY_CONNECTIVITY use new engine.

***** Start: VERIFY CONNECTIVITY *****
Start Time: Fri Dec 6 21:06:52 2024

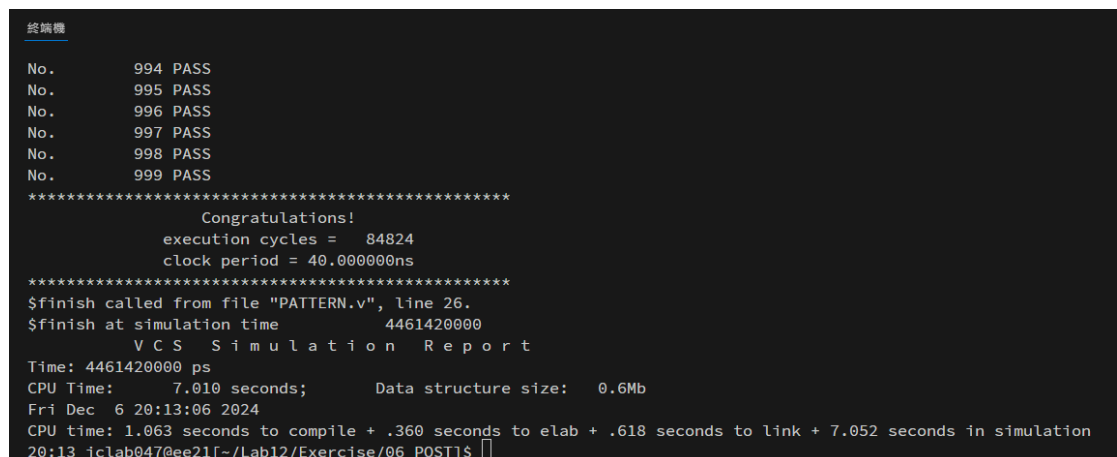
Design Name: CHIP
Database Units: 1000
Design Boundary: (0.0000, 0.0000) (2745.3600, 2745.3600)
Error Limit = 1000; Warning Limit = 50
Check all nets

Begin Summary
Found no problems or warnings.
End Summary

End Time: Fri Dec 6 21:06:53 2024
Time Elapsed: 0:00:01.0

***** End: VERIFY CONNECTIVITY *****
Verification Complete : 0 Viols. 0 Wrngs.
(CPU Time: 0:00:00.1 MEM: 13.000M)
```

7. Post Layout simulation result :

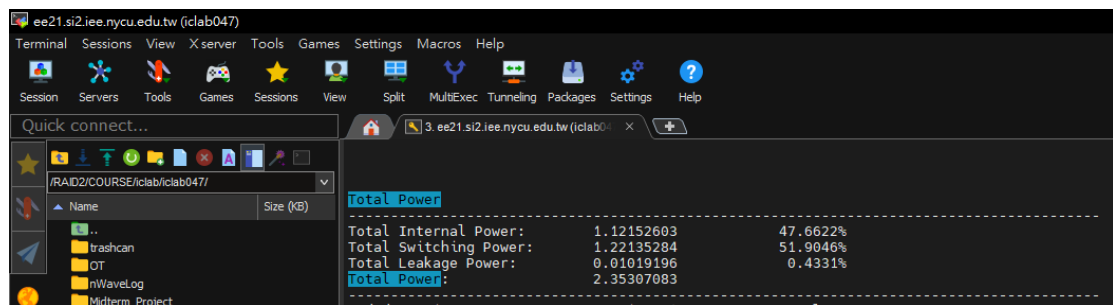


```
终端
No. 994 PASS
No. 995 PASS
No. 996 PASS
No. 997 PASS
No. 998 PASS
No. 999 PASS

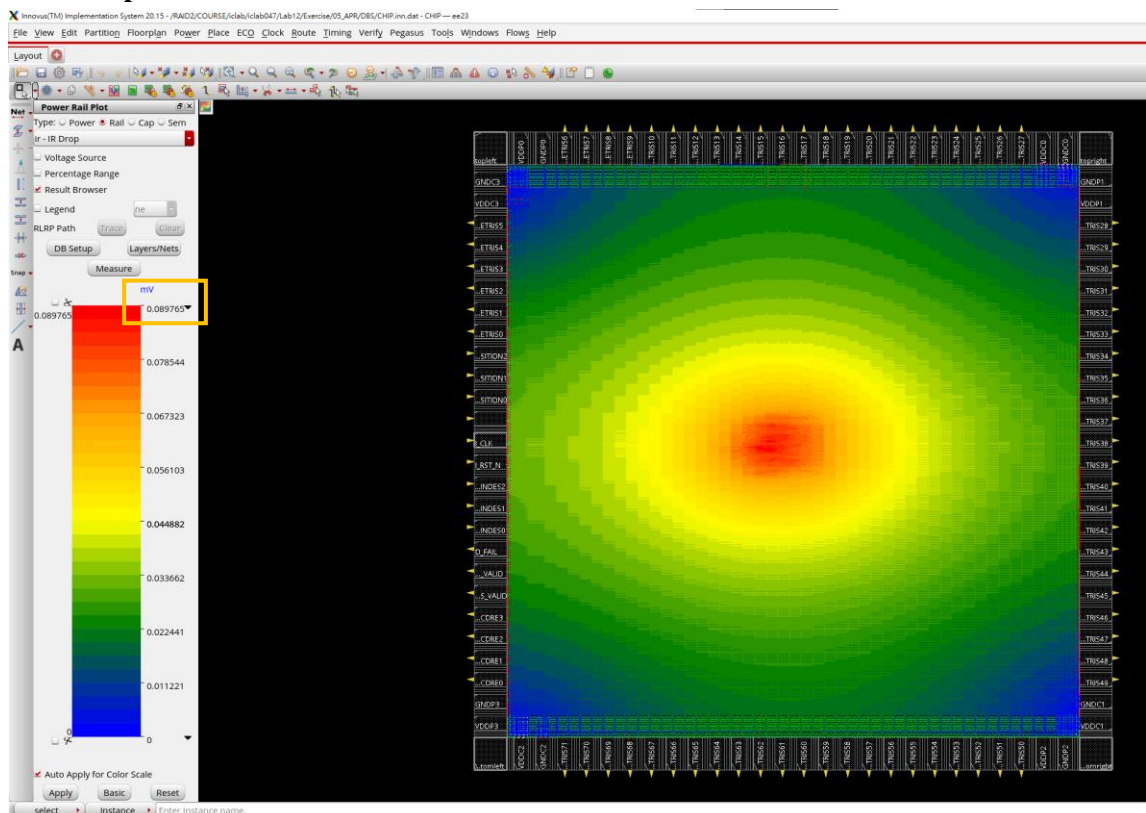
*****
Congratulations!
execution cycles = 84824
clock period = 40.000000ns

*****
$finish called from file "PATTERN.v", line 26.
$finish at simulation time 4461420000
VCS Simulation Report
Time: 4461420000 ps
CPU Time: 7.010 seconds; Data structure size: 0.6Mb
Fri Dec 6 20:13:06 2024
CPU time: 1.063 seconds to compile + .360 seconds to elab + .618 seconds to link + 7.052 seconds in simulation
20:13 iclab047@ee21[~/Lab12/Exercise/06_POST]$
```

8. Power result :



9. IR Drop Results :



如果 IR drop 過大，會導致晶片的運行速度降低，尤其在先進製程中，電子元件更容易受到 IR drop 的影響，IR drop 可能影響到 Performance (circuit running slower than specification)、Functionality problem (setup or hold violations)、Unreliable operation (less noisemargin)。

預防 IR drop 的方法有：

1. Adding stripes to avoid IR drop on cell's power line
2. Change the position of power pad to avoid long power line
3. Use square-shaped Chip to avoid long power line
4. Add decoupling cell

我採用了方法 1. Adding stripes to avoid IR drop on cell's power line，在 Lab11 中 Stripe 的間距為 100，但在本次實驗中為了降低 IR drop on cell's power line，我將間距調整為 50。